

BIRLA INSTITUTE OF TECHNOLOGY MESRA, RANCHI
DEPARTMENT OF MATHEMATICS

MA24103 Mathematics-II

Session: SP 2026

Tutorial - 1 (Module I)

1.
 - a) Show that the Wronskian of $e^{ax} \cos(bx)$ and $e^{ax} \sin(bx)$ where $b \neq 0$, is $b^2 e^{2ax}$.
 - b) Prove that the Wronskian of functions $e^{m_1 x}$, $e^{m_2 x}$, $e^{m_3 x}$ is equal to $(m_1 - m_2)(m_2 - m_3)(m_3 - m_1)e^{(m_1+m_2+m_3)x}$.
 - c) Show that $y_1(x) = \sin x$ and $y_2(x) = \sin x - \cos x$ are linearly independent solutions of $y'' + y = 0$.
 - d) Show that the Wronskian of the function x^2 , $x^2 \log(x)$ is non-zero. Can these functions be independent solution of an ordinary differential equation; if so determine this equation.
 - e) Check whether e^x , xe^x , $\sinh(x)$ are linearly independent or dependent.
 - f) Check whether e^{ikx} , e^{-ikx} , $\sin(kx)$ are linearly independent or not.
 - g) Show that any pair of these four functions are linearly independent: e^{ikx} , e^{-ikx} , $\sin(kx)$, $\cos(kx)$.
2. Solve the following differential equations:

a) $y'' + 3y' - 10y = 0$.	b) $y''' - 9y'' + 23y' - 15y = 0$.
c) $y^{iv} + 8y'' + 16y = 0$.	d) $y'' + 3y' + 4y = x^3 - 2x$.
e) $y'' - 3y' + 2y = e^{3x}$.	f) $y'' - 4y' + 4y = e^{2x} + x^3 + \cos(2x)$.
g) $y''' - 7y' - 6y = e^{2x}(1 + x)$.	h) $y''' - 3y'' + 4y' - 2y = e^x + \cos x$.
i) $y''' - 3y'' - y' + 3y = x^2 e^x$.	j) $y'' - 6y' + 9 = 1 + x + x^2$.
k) $y'' + 9y = x \cos x$.	
3. Solve the following Cauchy-Euler equations:

a) $y'' + \frac{1}{x}y' = 12 \log x/x^2$.	b) $x^2 y'' + 3xy' + y = 1/(1-x)^2$.
c) $x^2 y'' - 2y = 2x + 6$.	d) $x^2 y'' - 3xy' + 3y = 2 + 3 \ln x$.
e) $x^2 y'' + 2xy' = \cos(\ln x)$.	f) $x^2 y'' + 2xy' - 2y = x \sin(\ln x)$.
4. Solve by the method of variation of parameters

a) $y'' + y = x$.	b) $y'' + n^2 y = \sec(nx)$.
c) $y'' - y = \frac{2}{1 + e^x}$.	d) $y'' + y = \tan x$.
e) $y'' - 2y' + y = e^x$.	f) $y'' + 4y = x \sin 2x$.